A Lattice-Theoretic Abstraction of PDR via Adjunctions

Mayuko Kori

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based on [Kori+, CAV'22], [Kori+, CAV'23], [Kori+, FMSD'26]
joint work with Natsuki Urabe, Shin-ya Katsumata, Kohei Suenaga, Ichiro Hasuo, Flavio Ascari, Filippo Bonchi, Roberto Bruni, Roberta Gori

Property Directed Reachability (PDR)

- Originally: [Bradley, VMCAI'11], [Een+, FMCAD'11]
  an effective SAT-based algorithm for safety model checking
- Input: A transition system and a safety property
  represented by formulas

$$I(x), T(x, x'), P(x)$$

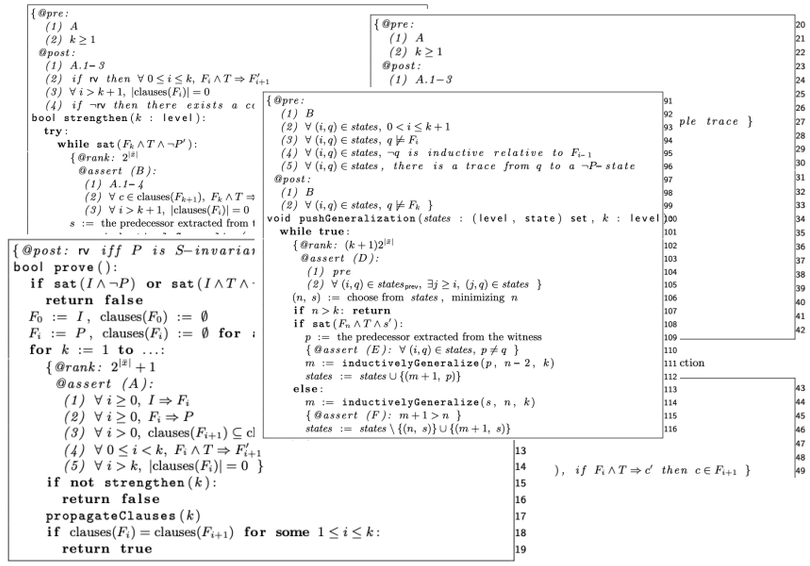
initial | transition $x \to x'$ | Safe

- Target problem: Are all reachable states safe?
- Output: Yes, with an **invariant** / No, with a **counterexample**

interaction between two searches

PDR-style Algorithms

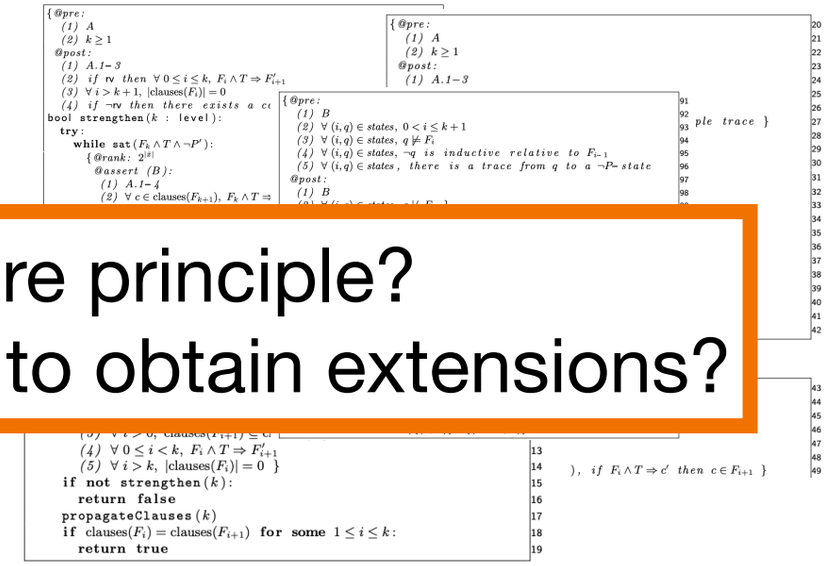
- Highly effective, especially in hardware model checking
- Many extensions and variants:
  - software model checking, CHC solving [CimattiG, CAV'12], [HoderB SAT'12]
  - probabilistic systems [Batz+, CAV'20],
  - higher-order systems [Katsura+, ICFP'23],
  - petri nets [Amat+, TACAS'22], …
- They are often highly heuristic-dependent



*[Bradley, VMCAI'11]

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a core principle?
a unified way to obtain extensions?

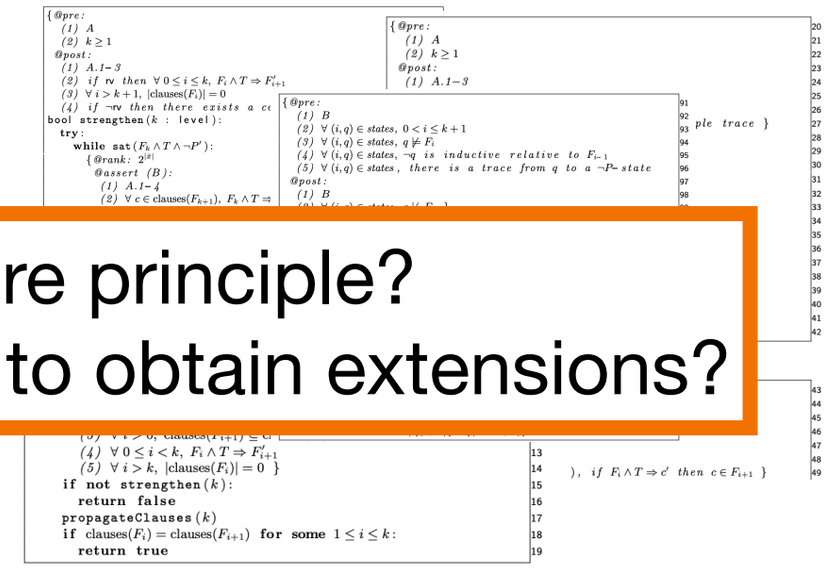
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*[Bradley, VMCAI'11]

Background

PDR-style Algorithms

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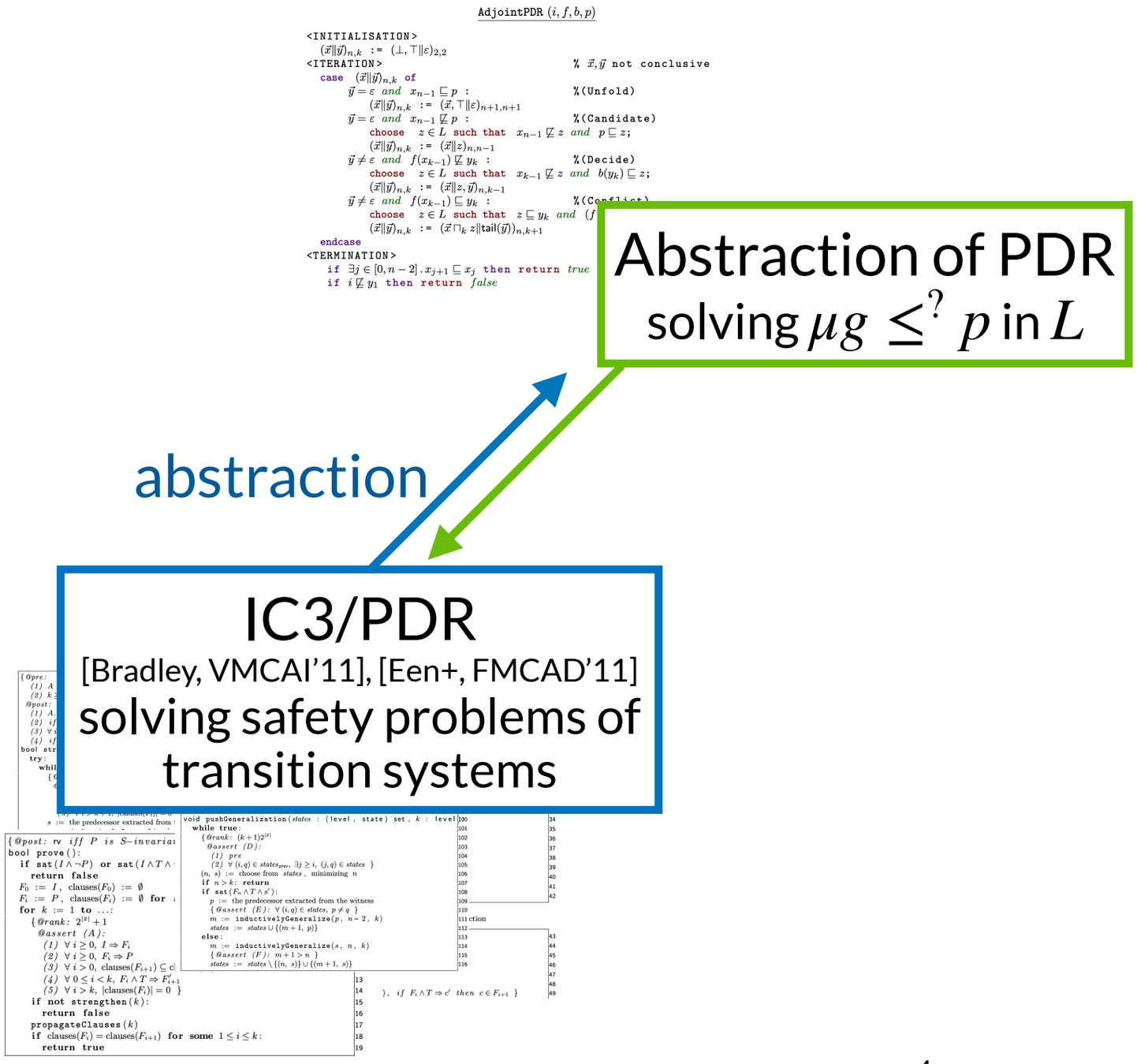
a core principle?
a unified way to obtain extensions?

- They are often highly heuristic-dependent *[Bradley, VMCAI'11]

any guide to design heuristics?

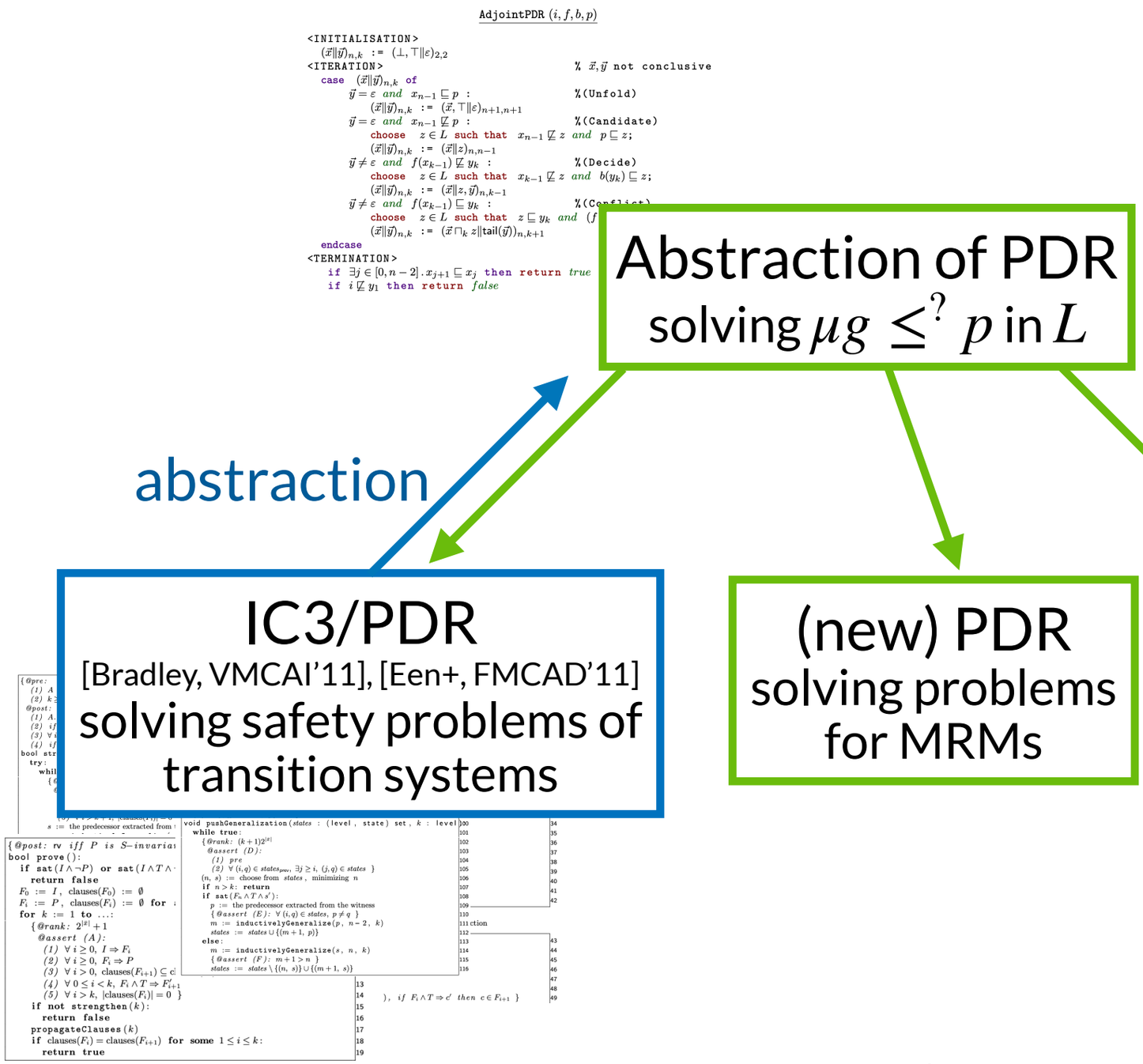
Our work in [Kori+, CAV'22, CAV'23]

Lattice-theoretic abstraction of PDR



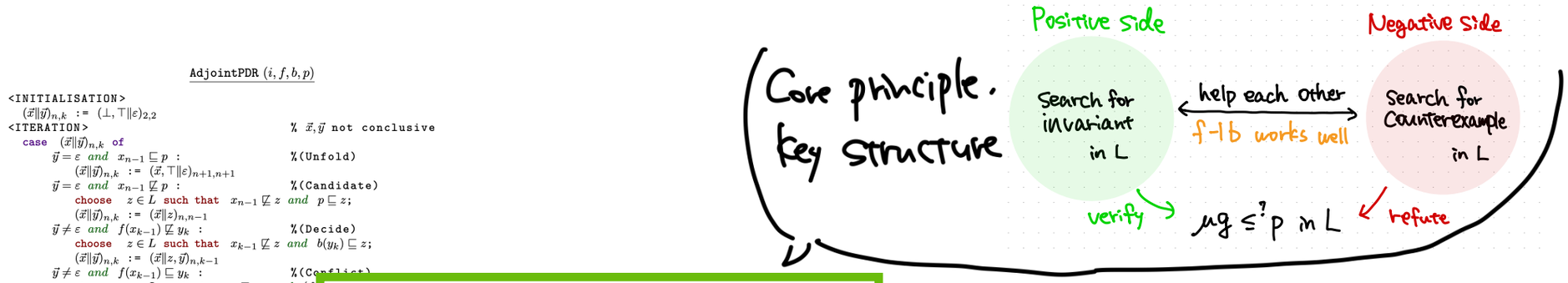
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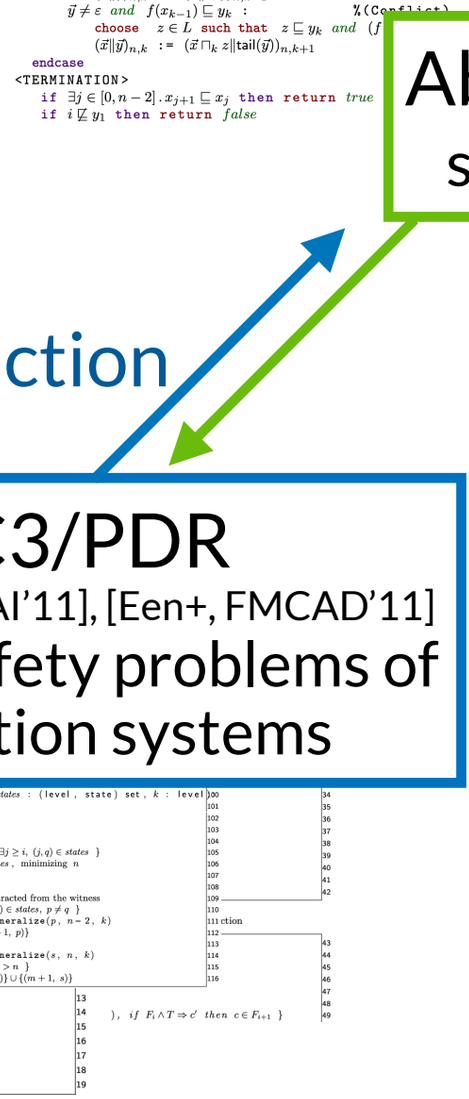
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Lattice-theoretic abstraction of PDR





Abstraction of PDR
solving $\mu g \leq^{?} p$ in L

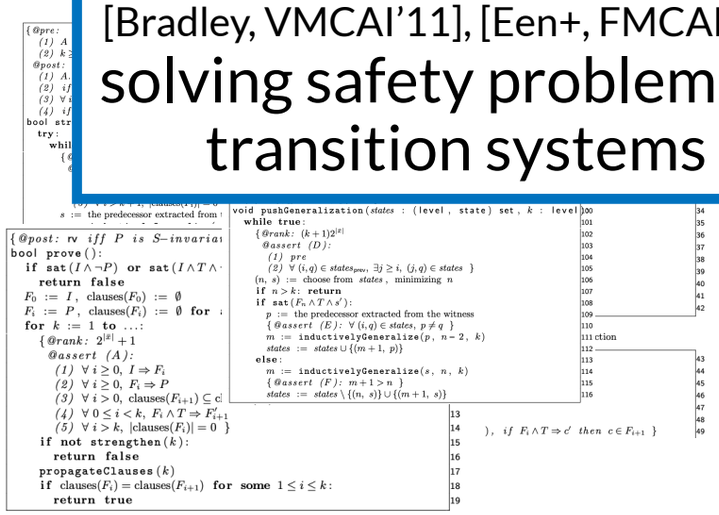
abstraction

instantiation
by specifying $L, g : L \to L, \ldots$

IC3/PDR
[Bradley, VMCAI'11], [Een+, FMCAD'11]
solving safety problems of transition systems

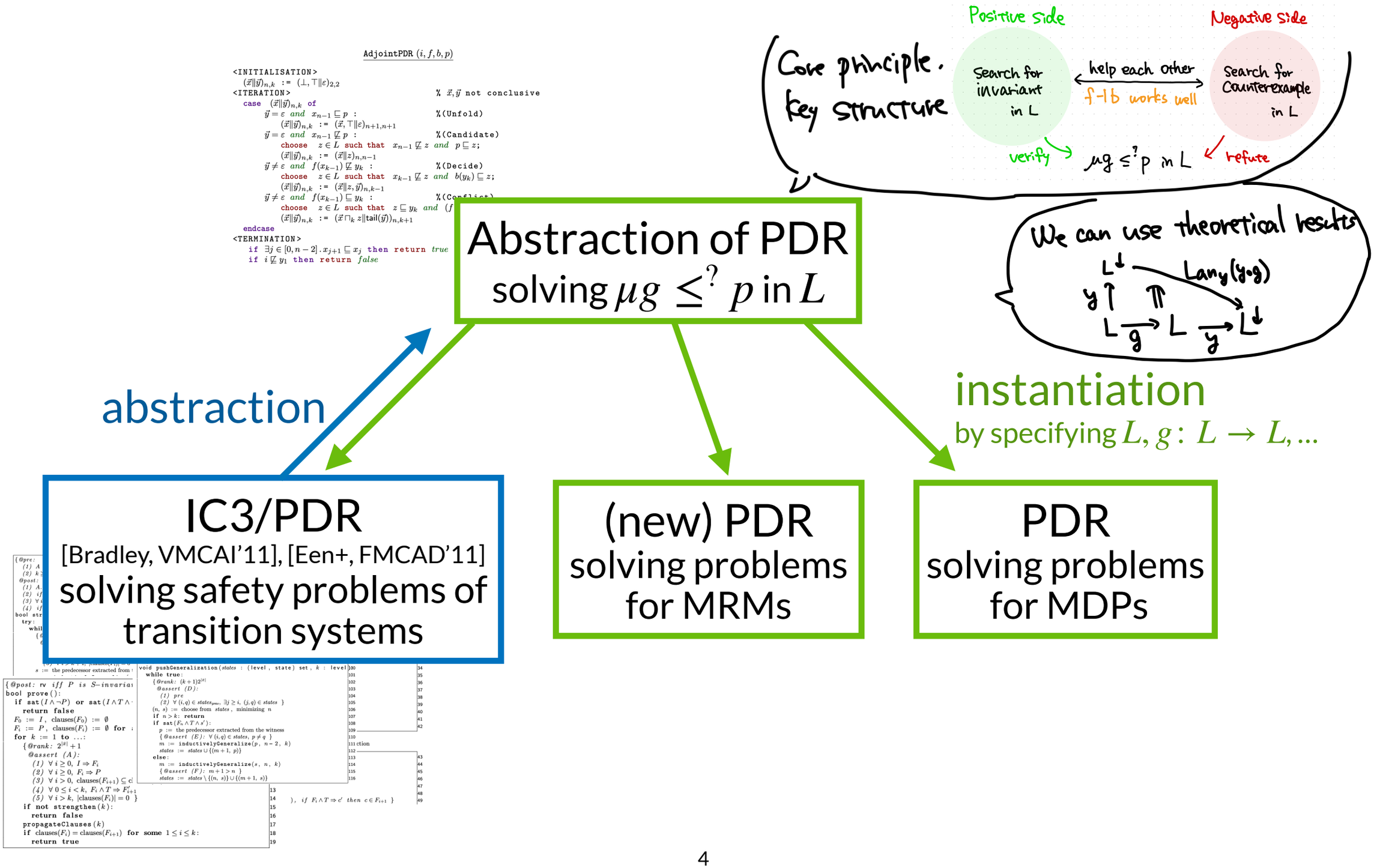
(new) PDR
solving problems for MRMs

PDR
solving problems for MDPs



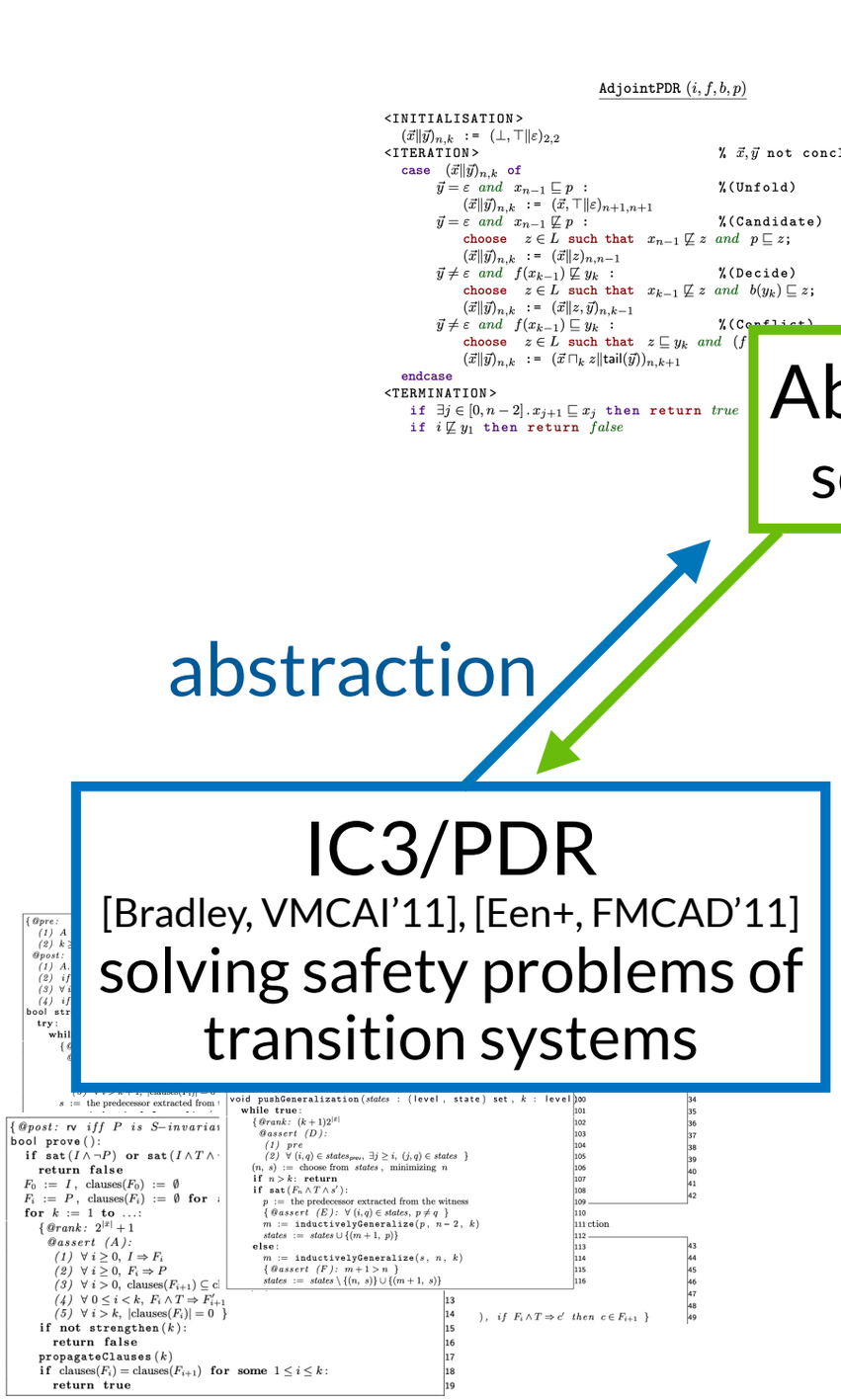
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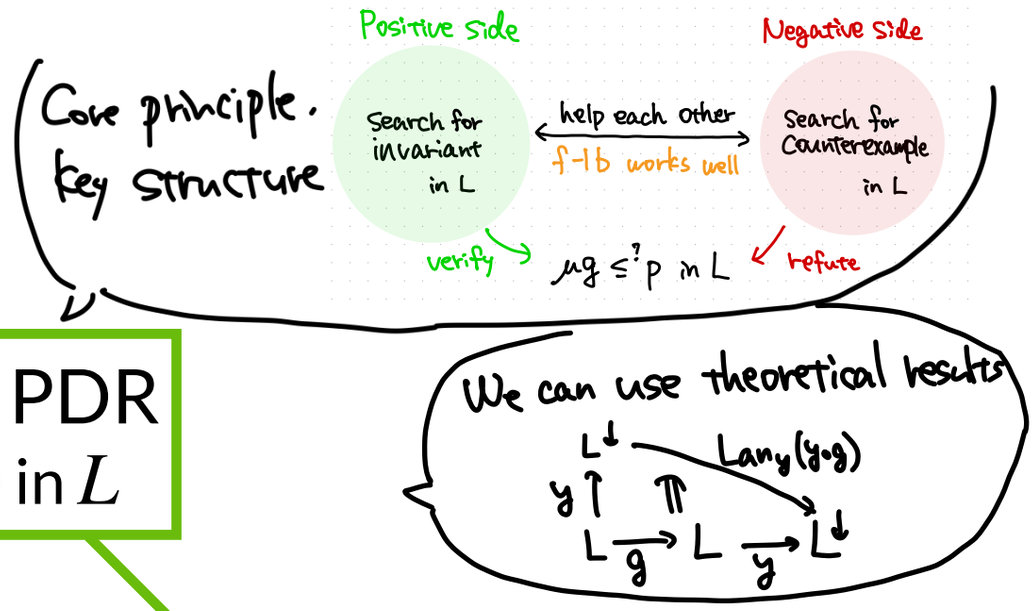
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Lattice-theoretic abstraction of PDR





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Theory-guided heuristics

Benchmark	$\|S\|$	P	λ	AdjointPDR↓ hCoB	AdjointPDR↓ hCo01	AdjointPDR↓ hCoS	LT-PDR	PrIC3 none	PrIC3 lin.	PrIC3 pol.	PrIC3 hyb.
Grid	10^2	0.033	0.3	0.013	0.022	0.659	0.343	1.383	23.301	MO	MO
			0.2	0.013	0.031	0.657	0.519	1.571	26.668	TO	MO
	10^3	<0.001	0.3	1.156	2.187	5.633	126.441	TO	TO	TO	MO
			0.2	1.146	2.133	5.632	161.667	TO	TO	TO	MO
BRP	10^3	0.035	0.1	12.909	7.969	55.788	TO	TO	TO	MO	MO
			0.01	1.977	8.111	5.645	21.078	60.738	626.052	524.373	823.082
			0.005	0.604	2.261	2.709	1.429	12.171	254.000	197.940	318.840
Zero-Conf	10^2	0.5	0.9	1.217	68.937	0.196	TO	19.765	136.491	0.630	0.468
			0.75	1.223	68.394	0.636	TO	19.782	132.780	0.602	0.467
			0.52	1.228	60.024	0.739	TO	19.852	136.533	0.608	0.474
			0.45	<0.001	0.001	0.001	<0.001	0.035	0.043	0.043	0.043
	10^4	0.5	0.9	MO	TO	7.443	TO	TO	TO	0.602	0.465
			0.75	MO	TO	15.223	TO	TO	TO	0.599	0.470
			0.52	MO	TO	TO	TO	TO	TO	0.488	0.475

Outline

1. Lattice-theoretic abstraction of safety problems

$$\mu g \leq^{?} p \text{ in } L \text{ (with } f \dashv b\text{)}$$

2. AdjointPDR for $\mu g \leq^{?} p$ with $f \dashv b \colon L \to L$
3. How we recover the key structure $f \dashv b$ for general $\mu g \leq^{?} p$
4. Heuristics (for MDPs)

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Abstraction of Safety Problems

Def. Lattice-theoretic abstraction of safety problems:

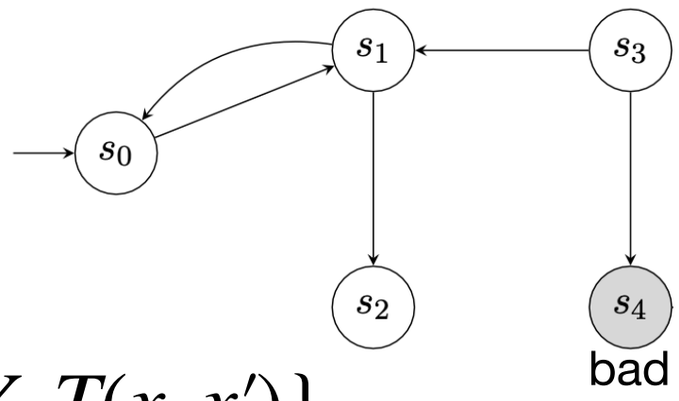
$$\mu g \leq^{?} p \text{ in } L$$

where L is a complete lattice, $g : L \to L$ is ω-conti., $p \in L$
μg is the least fixed point of g.

e.g. transition system.

I: initial states, P: safe states,

T: the transition relation.



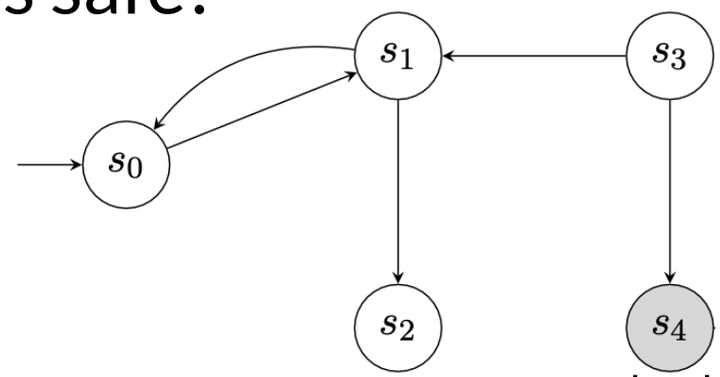
- $L := (\mathcal{P}S, \subseteq), g(X) := I \cup \{x' \mid x \in X, T(x, x')\}$
- μg = the set of reachable states.

$\mu g \leq P$ iff all reachable states are safe.

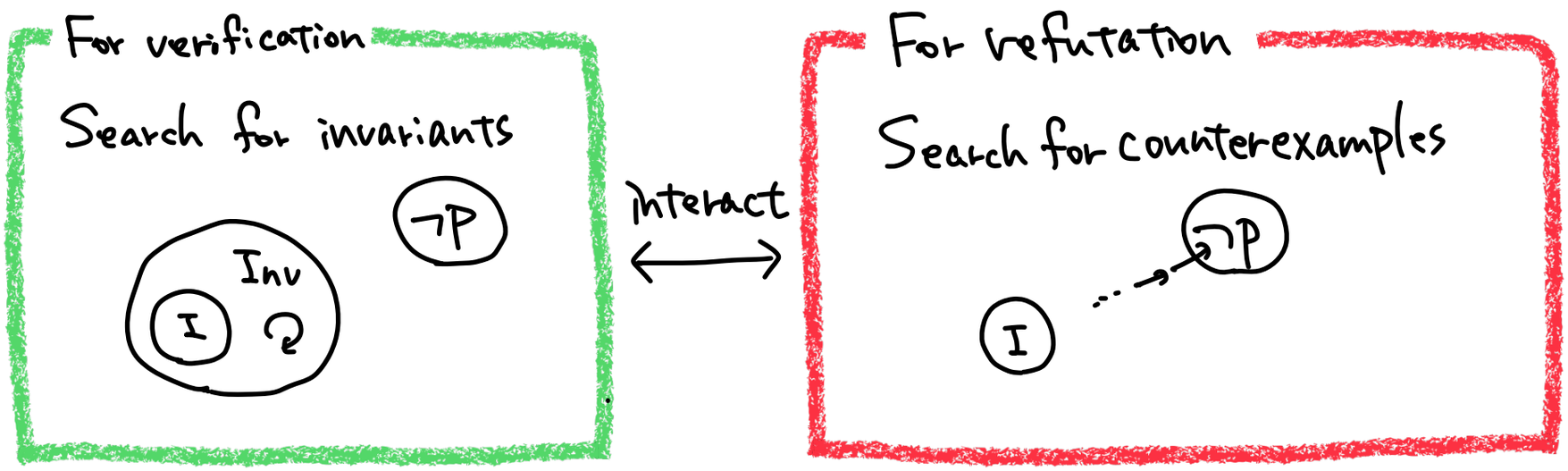
Abstraction of witnesses

Given (I, T, P), are all reachable states safe?

I: initial states, P: safe states
T: the transition relation.



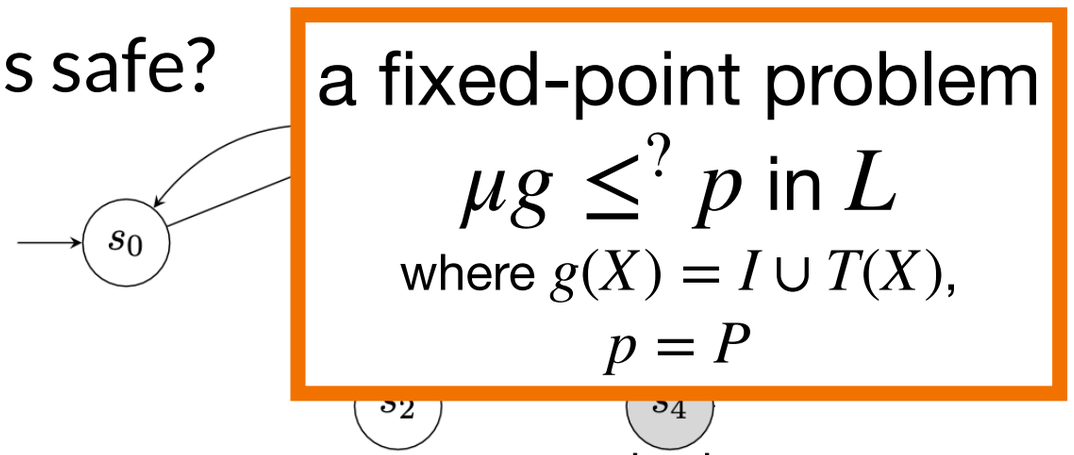
PDR searches for two kinds of witness **at the same time**



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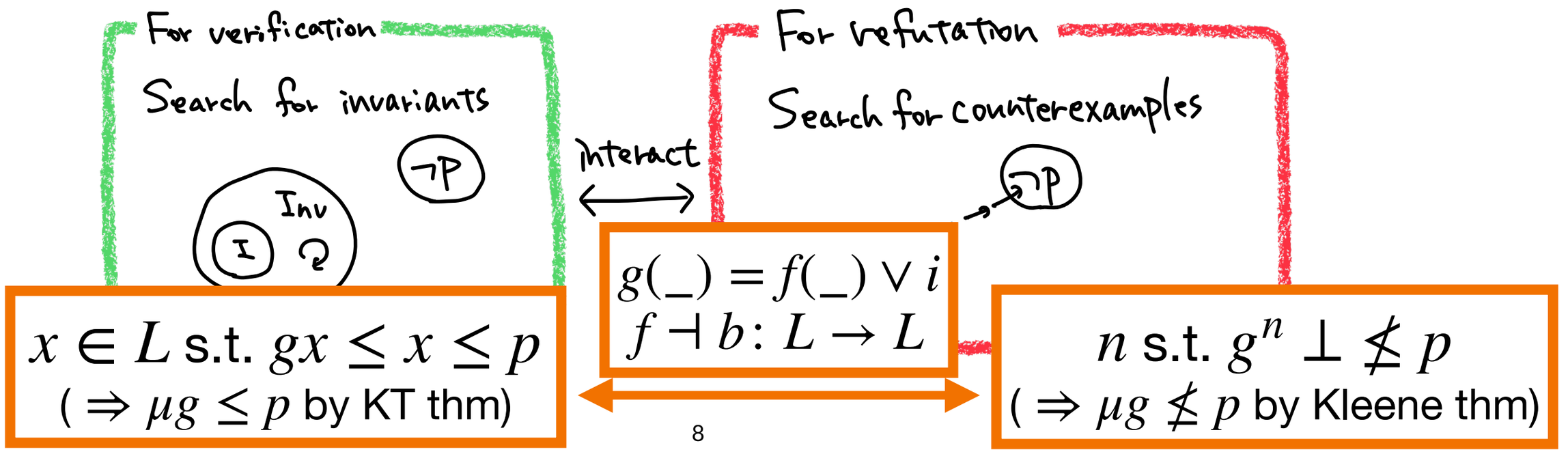


a fixed-point problem

$\mu g \leq^{?} p$ in L

where $g(X) = I \cup T(X)$,
$p = P$

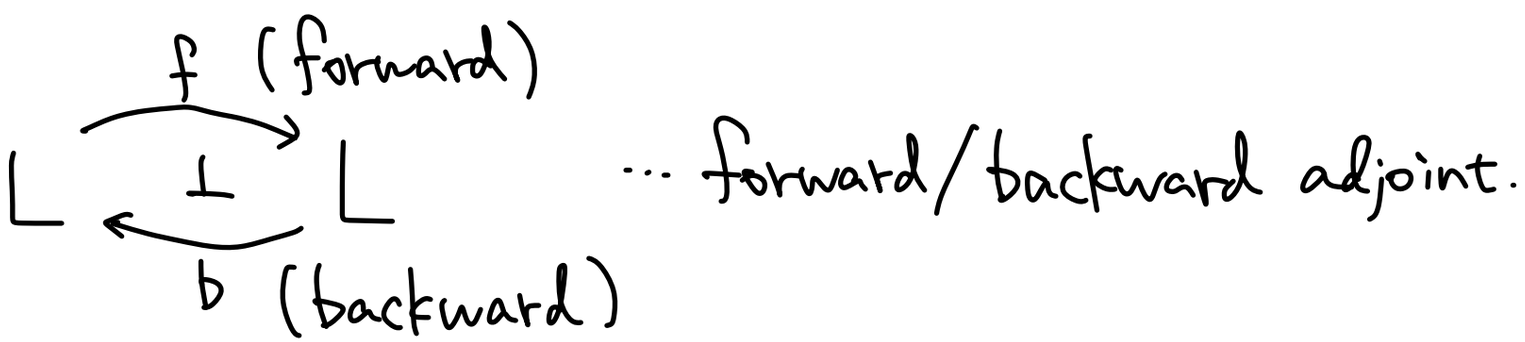
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key structure $f \dashv b$ for the interaction

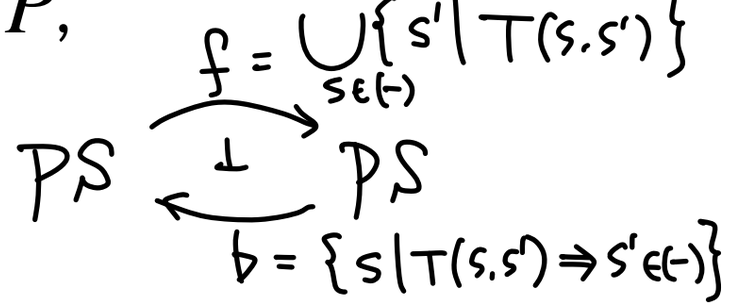
Target Safety Problems for AdjointPDR

$\mu g \leq^{?} p$ in a complete lattice L
where g has the form $g(x) = f(x) \vee i$ $(i, p \in L, f \dashv b \colon L \to L)$



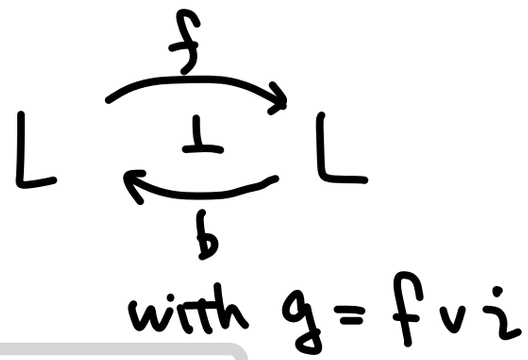
e.g. Safety problem of a transition system (I, T, P)

- $L := \mathscr{P}S$, $g(X) = \bigcup_{s \in X} \{s' \mid T(s, s')\} \cup I$, $p := P$,
- $\mu g \leq p$ iff all reachable states are safe.



key structure $f \dashv b$ for the interaction

Two Equivalent Forms of $\mu g \leq^{?} p$ with $f \dashv b$



Target problem

$$\mu g \leq p \Leftrightarrow \mu(f \vee i) \leq p$$

Forward form
(e.g. all reachable states are safe)

$$\Leftrightarrow \exists x\,.\, fx \vee i \leq x \leq p \qquad \text{(by KT thm)}$$

$$\Leftrightarrow \exists x\,.\, fx \leq x \text{ and } i \leq x \leq p$$

$$\Leftrightarrow \exists x\,.\, x \leq bx \text{ and } i \leq x \leq p \qquad \text{(by } f \dashv b\text{)}$$

$$\Leftrightarrow \exists x\,.\, i \leq x \leq bx \wedge p$$

$$\Leftrightarrow i \leq \nu(b \wedge p)$$

Backward form
(e.g. all initial states are safe after any steps)

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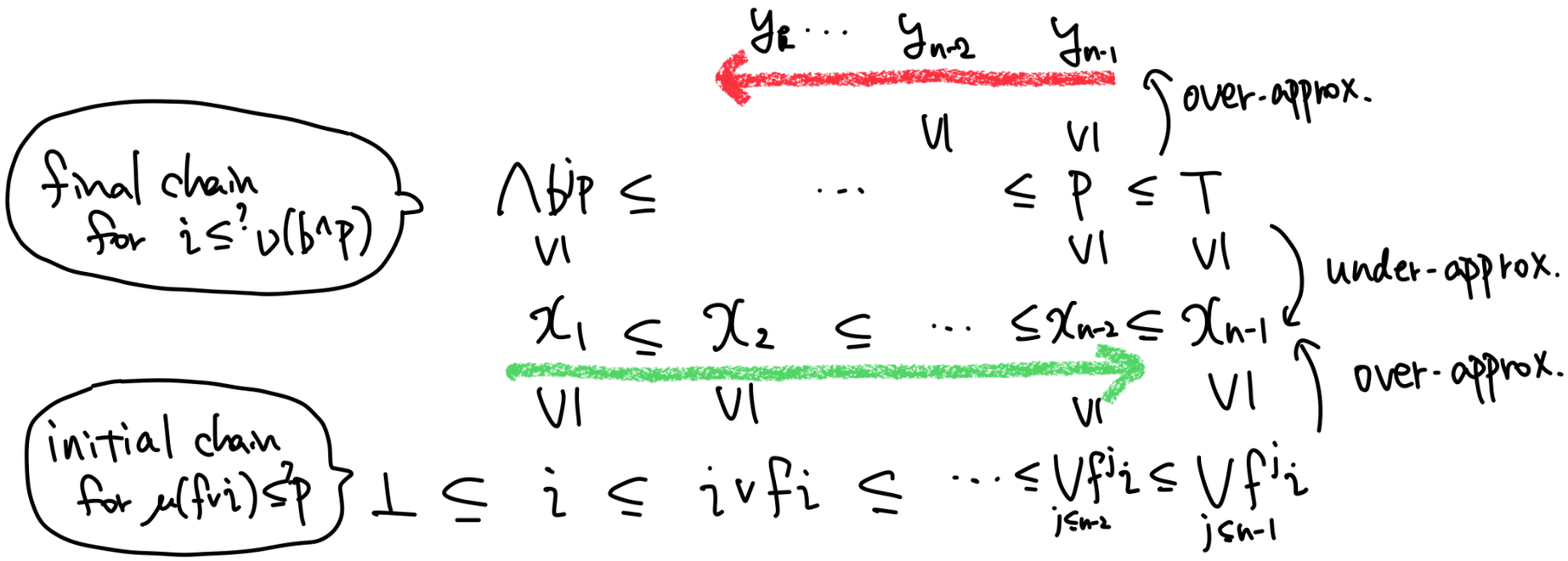
Lattice-theoretic abstraction of PDR

AdjointPDR[Kori+, CAV'23]

It solves $\mu g \leq^? p$ ($\Leftrightarrow \mu(f \vee i) \leq^? p \Leftrightarrow i \leq^? \nu(b \wedge p)$)

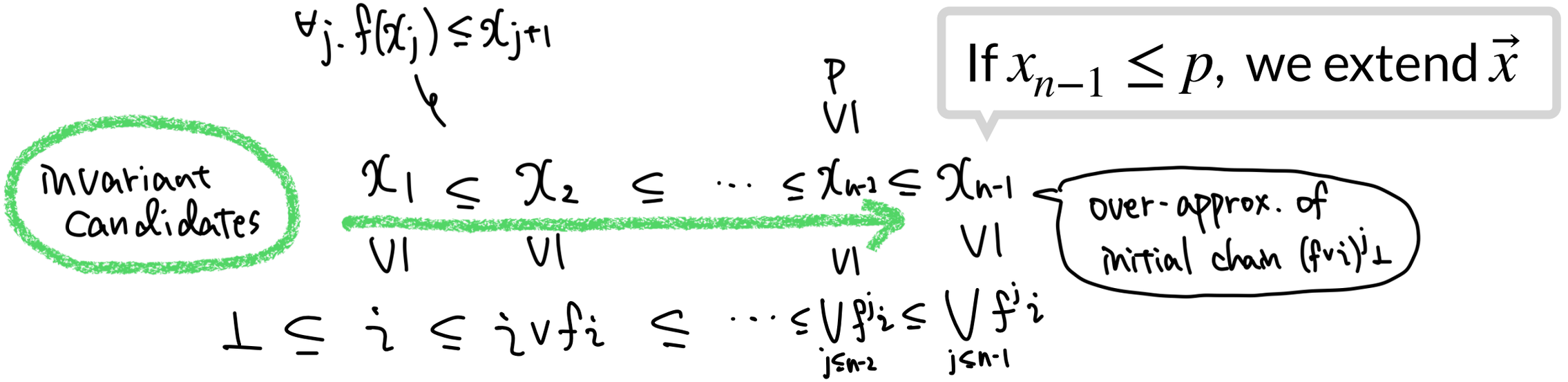
It consists of two side:

- Positive side, aiming to show $\mu(f \vee i) \leq p$
- Negative side, aiming to show $i \not\leq \nu(b \wedge p)$



Positive Side to show $\mu(f \vee i) \leq p$

Intermediate data: a positive chain $\vec{x} = (x_1 \leq \cdots \leq x_{n-1})$



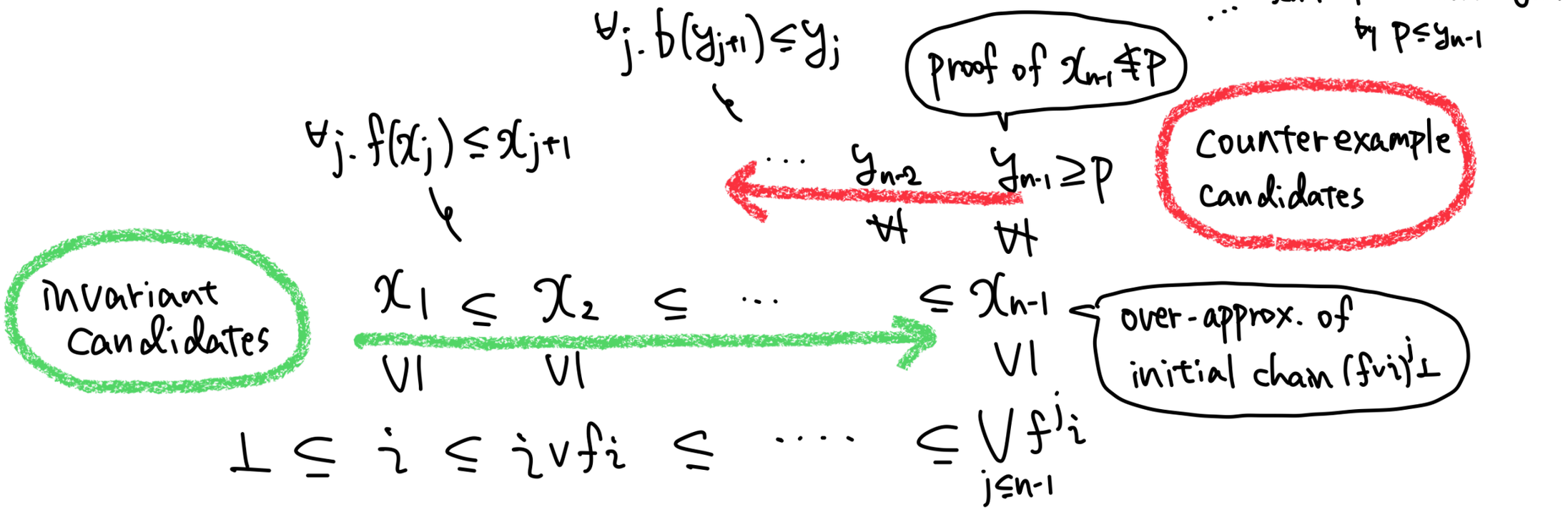
- Invariant candidate: If $\exists j\,.\ x_j = x_{j+1}$, then x_j is an invariant $\rightarrow \mu(f \vee i) \leq p$
- A proof of $(n-2)$-step safety ($\bigvee_{j \leq n-2} f^j i \leq p$)

Milestone: construct a longer $\vec{x}$

Intermediate data $(\vec{x}\|\vec{y})_{n,k}$ of AdjointPDR

Negative side to show $i \not\leq \nu(b \wedge p)$

Intermediate data: negative sequence $\vec{y} = (y_k, \cdots, y_{n-1})$



- Each y_j is a proof of unsafety of x_j (i.e. $x_j \not\leq b^{n-1-j}p$)

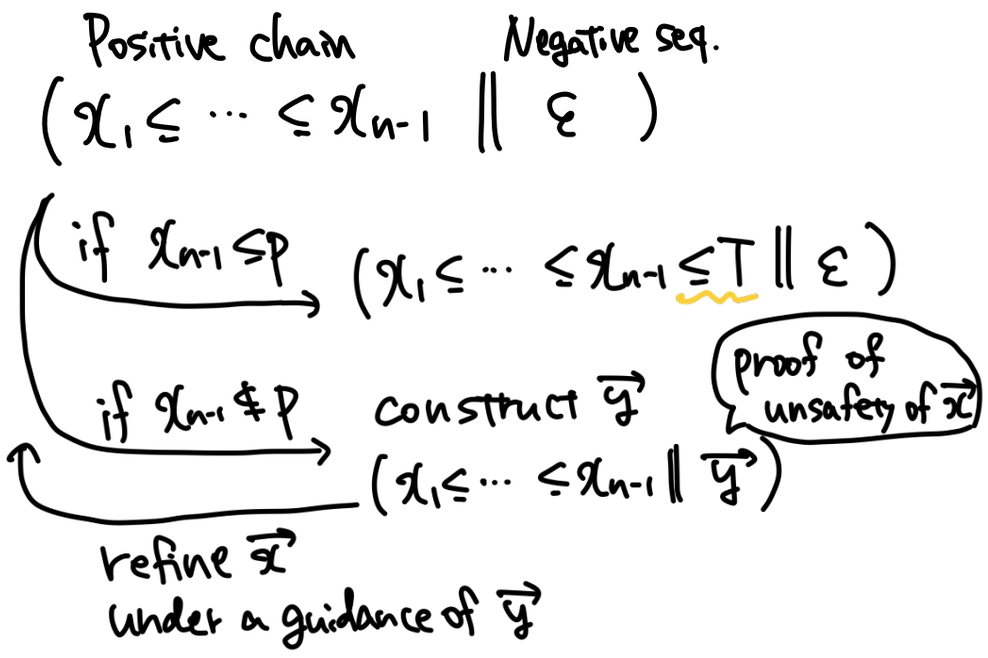
$\vec{y}$ is a guide to refine $\vec{x}$

- Counterexample candidates: If $i \not\leq y_1$, then $i \not\leq \nu(b \wedge p)$ holds

AdjointPDR for $\mu g \leq^{?} p$ with $f \dashv b$

AdjointPDR (i, f, b, p)

```
<INITIALISATION>
  (x⃗‖y⃗)_{n,k}  :=  (⊥, ⊤‖ε)_{2,2}
<ITERATION>                                          % x⃗, y⃗ not conclusive
  case  (x⃗‖y⃗)_{n,k}  of
      y⃗ = ε  and  x_{n-1} ⊑ p  :                      %(Unfold)
          (x⃗‖y⃗)_{n,k}  :=  (x⃗, ⊤‖ε)_{n+1,n+1}
      y⃗ = ε  and  x_{n-1} ⋢ p  :                      %(Candidate)
          choose  z ∈ L  such that  x_{n-1} ⋢ z  and  p ⊑ z;
          (x⃗‖y⃗)_{n,k}  :=  (x⃗‖z)_{n,n-1}
      y⃗ ≠ ε  and  f(x_{k-1}) ⋢ y_k  :                 %(Decide)
          choose  z ∈ L  such that  x_{k-1} ⋢ z  and  b(y_k) ⊑ z;
          (x⃗‖y⃗)_{n,k}  :=  (x⃗‖z, y⃗)_{n,k-1}
      y⃗ ≠ ε  and  f(x_{k-1}) ⊑ y_k  :                 %(Conflict)
          choose  z ∈ L  such that  z ⊑ y_k  and  (f ⊔ i)(x_{k-1} ⊓ z) ⊑ z;
          (x⃗‖y⃗)_{n,k}  :=  (x⃗ ⊓_k z‖tail(y⃗))_{n,k+1}
  endcase
<TERMINATION>
    if  ∃j ∈ [0, n−2] . x_{j+1} ⊑ x_j  then return true   % x⃗ conclusive
    if  i ⋢ y_1  then return false                        % y⃗ conclusive
```



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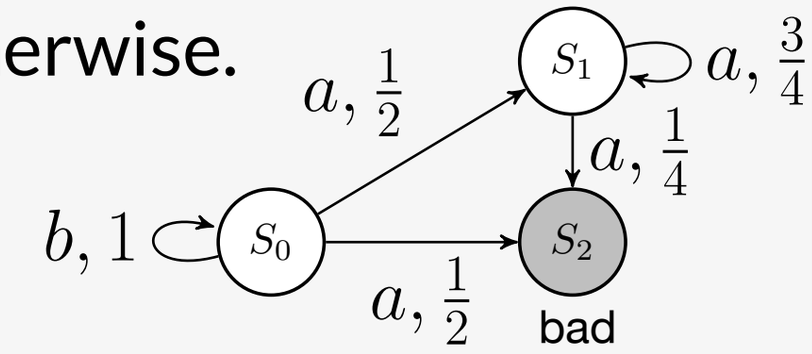
Lattice-theoretic abstraction of safety problems

Max reachability Problems of MDPs

e.g. (MDP $(S, A, \delta\colon S \times A \to \mathscr{D}S + 1)$, bad states $B \subseteq S$)

- $L := [0,1]^S, i(s) = 1$ if $s \in B, i(s) = 0$ otherwise.

  $$f(d)(s) := \max_a \sum_{s'} d(s') \cdot \delta(s, a, s')$$

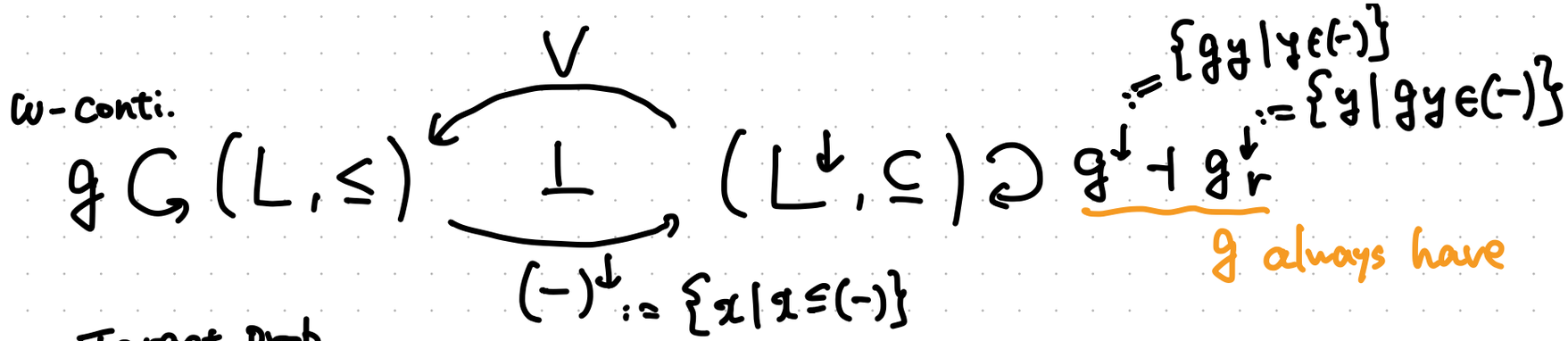


- $(\mu(f \vee i))(s) = \Pr^{\max}(s \vDash \Diamond B)$
- $\mu(f \vee i) \leq \begin{pmatrix} s_0 \mapsto \lambda \\ _ \mapsto 1 \end{pmatrix}$ in $[0,1]^S \Leftrightarrow \Pr^{\max}(s_0 \vDash \Diamond B) \leq \lambda$ in $[0,1]$

The Bellman operator f has no right adjoint.

Thus this problem is outside the scope of AdjointPDR.

Recovering Adjoints by Lower Sets



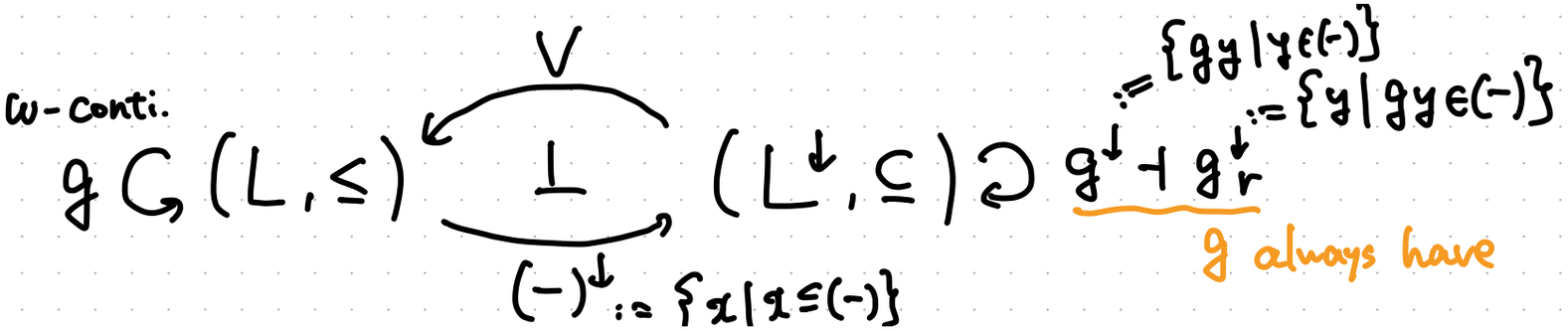
- $L^{\downarrow}$ is too large to get a convergence
- AdjointPDR$^{\downarrow}$: positive chain in L,
  negative sequence in $L^{\downarrow}$

AdjointPDR$^{\downarrow}$ (g, p)

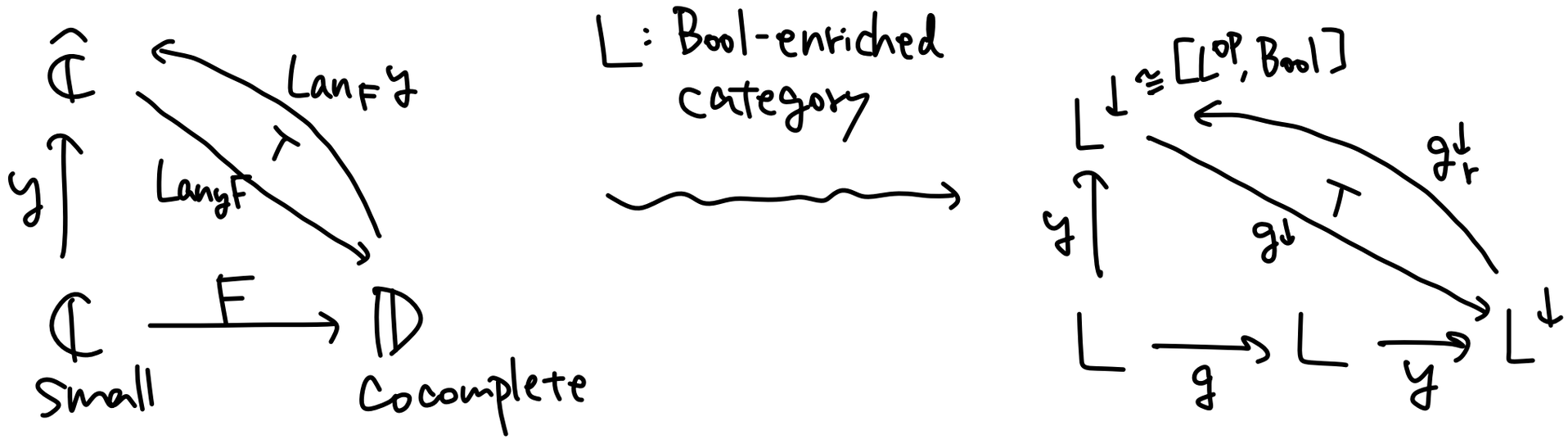
<INITIALISATION>
  $(\boldsymbol{x} \| \boldsymbol{Y})_{n,k} := (\emptyset, \bot, \top \| \varepsilon)_{3,3}$
<ITERATION>
  case $(\boldsymbol{x} \| \boldsymbol{Y})_{n,k}$ of % $\boldsymbol{x}, \boldsymbol{Y}$ not conclusive
    $\boldsymbol{Y} = \varepsilon$ *and* $x_{n-1} \sqsubseteq p$: %(Unfold)
      $(\boldsymbol{x} \| \boldsymbol{Y})_{n,k} := (\boldsymbol{x}, \top \| \varepsilon)_{n+1,n+1}$
    $\boldsymbol{Y} = \varepsilon$ *and* $x_{n-1} \not\sqsubseteq p$: %(Candidate)
      choose $Z \in L^{\downarrow}$ such that $x_{n-1} \notin Z$ *and* $p \in Z$;
      $(\boldsymbol{x} \| \boldsymbol{Y})_{n,k} := (\boldsymbol{x} \| Z)_{n,n-1}$
    $\boldsymbol{Y} \neq \varepsilon$ *and* $g(x_{k-1}) \notin Y_k$: %(Decide)
      choose $Z \in L^{\downarrow}$ such that $x_{k-1} \notin Z$ *and* $g_r^{\downarrow}(Y_k) \subseteq Z$;
      $(\boldsymbol{x} \| \boldsymbol{Y})_{n,k} := (\boldsymbol{x} \| Z, \boldsymbol{Y})_{n,k-1}$
    $\boldsymbol{Y} \neq \varepsilon$ *and* $g(x_{k-1}) \in Y_k$: %(Conflict)
      choose $z \in L$ such that $z \in Y_k$ *and* $g(x_{k-1} \sqcap z) \sqsubseteq z$;
      $(\boldsymbol{x} \| \boldsymbol{Y})_{n,k} := (\boldsymbol{x} \sqcap_k z \| \mathsf{tail}(\boldsymbol{Y}))_{n,k+1}$
  endcase
<TERMINATION>
  if $\exists j \in [0, n-2] \,.\, x_{j+1} \sqsubseteq x_j$ then return *true* % $\boldsymbol{x}$ conclusive
  if $Y_1 = \emptyset$ then return *false* % $\boldsymbol{Y}$ conclusive

Remark | Yoneda Cocompletion

The lower-set construction



can be seen as the Yoneda cocompletion of the poset L.



Outline

1. Lattice-theoretic abstraction of safety problems

$$\mu g \leq^{?} p \text{ in } L \text{ (with } f \dashv b\text{)}$$

2. AdjointPDR for $\mu g \leq^{?} p$ with $f \dashv b : L \to L$
3. How we recover the key structure $f \dashv b$ for general $\mu g \leq^{?} p$
4. Heuristics (for MDPs)

Heuristics for AdjointPDR$^{\downarrow}$

1. How to construct negative seq. Y

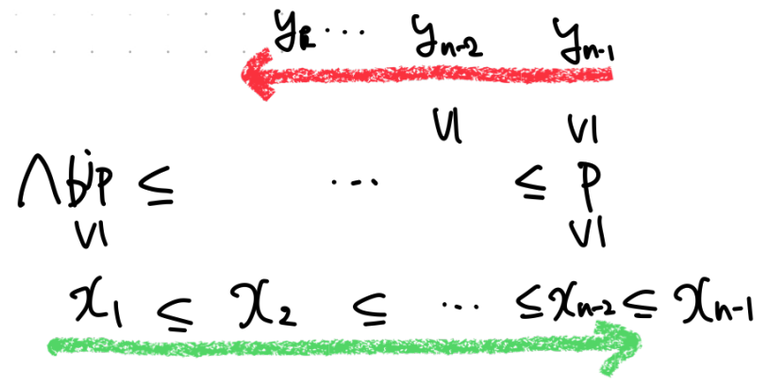
2. How to shrink overly-inflated positive chain x.

AdjointPDR$^{\downarrow}$ (g, p)

<INITIALISATION>
  $(\boldsymbol{x}\|\boldsymbol{Y})_{n,k}$:= $(\emptyset, \bot, \top\|\varepsilon)_{3,3}$
<ITERATION>
  case $(\boldsymbol{x}\|\boldsymbol{Y})_{n,k}$ of % $\boldsymbol{x}, \boldsymbol{Y}$ not conclusive
    $\boldsymbol{Y} = \varepsilon$ *and* $x_{n-1} \sqsubseteq p$: %(Unfold)
      $(\boldsymbol{x}\|\boldsymbol{Y})_{n,k}$:= $(\boldsymbol{x}, \top\|\varepsilon)_{n+1,n+1}$
    $\boldsymbol{Y} = \varepsilon$ *and* $x_{n-1} \not\sqsubseteq p$: %(Candidate)
      choose $Z \in L^{\downarrow}$ such that $x_{n-1} \notin Z$ *and* $p \in Z$;
      $(\boldsymbol{x}\|\boldsymbol{Y})_{n,k}$:= $(\boldsymbol{x}\|Z)_{n,n-1}$
    $\boldsymbol{Y} \neq \varepsilon$ *and* $g(x_{k-1}) \notin Y_k$: %(Decide)
      choose $Z \in L^{\downarrow}$ such that $x_{k-1} \notin Z$ *and* $g_r^{\downarrow}(Y_k) \subseteq Z$;
      $(\boldsymbol{x}\|\boldsymbol{Y})_{n,k}$:= $(\boldsymbol{x}\|Z, \boldsymbol{Y})_{n,k-1}$
    $\boldsymbol{Y} \neq \varepsilon$ *and* $g(x_{k-1}) \in Y_k$: %(Conflict)
      choose $z \in L$ such that $z \in Y_k$ *and* $g(x_{k-1} \sqcap z) \sqsubseteq z$;
      $(\boldsymbol{x}\|\boldsymbol{Y})_{n,k}$:= $(\boldsymbol{x} \sqcap_k z\|\mathsf{tail}(\boldsymbol{Y}))_{n,k+1}$
  endcase
<TERMINATION>
  if $\exists j \in [0, n-2]\,.\, x_{j+1} \sqsubseteq x_j$ then return *true* % $\boldsymbol{x}$ conclusive
  if $Y_1 = \emptyset$ then return *false* % $\boldsymbol{Y}$ conclusive

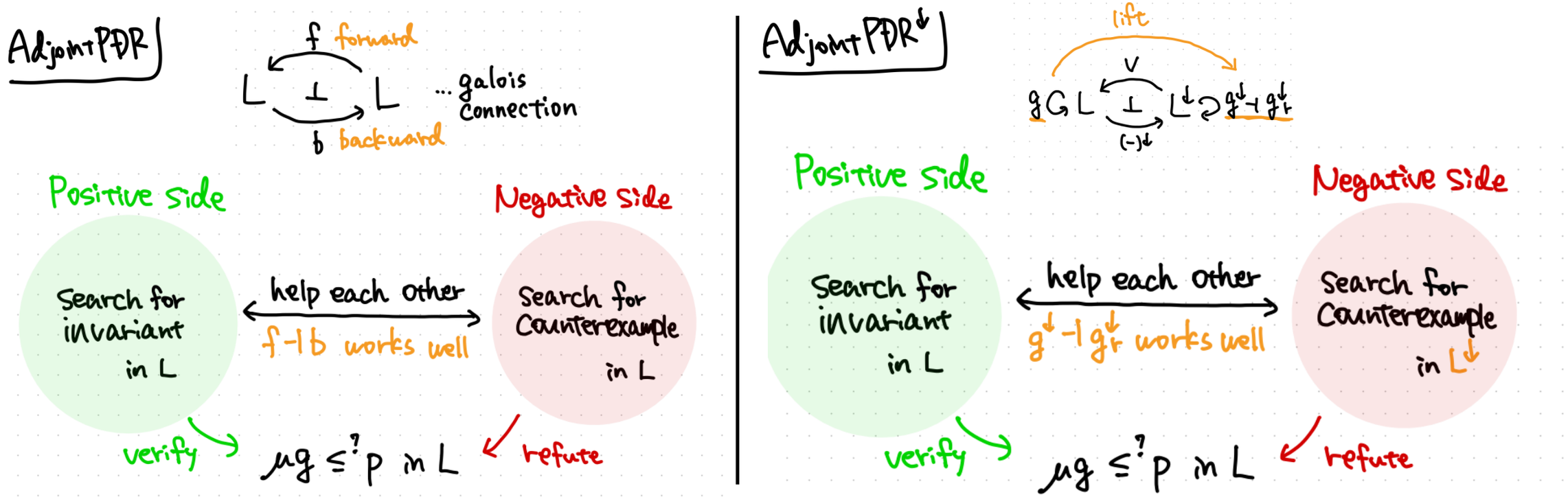
A canonical heuristics:

$$\overrightarrow{Y} = (g_r^{\downarrow n-k} p^{\downarrow}, \cdots, g_r^{\downarrow} p^{\downarrow}, p^{\downarrow}).$$



Based on it, we developed a practical heuristics for MDPs

Summary



These abstractions give:

- a unified framework for PDR-style extensions,
- identification and recovery of the key structure ($f \dashv b$),
- guidance for heuristic design.